MCBB 3.0 Proposed Direct Connect design Architecture

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1. Overview

This document is the comprehensive reference the agreed Direct Connect Gateway (DXGW) design pattern for MCBB 3.0 Cloud WAN deployment. This design pattern highlights Regional DXGW architecture with geographic segmentation: North American Edge Connects locations establish redundant Direct Connect circuits to their regional DXGW, with each NA DXGW associated to all North American CNEs for full mesh connectivity. International regions (Europe, Asia-Pacific, Latin America, Middle East) connect each data center to its geographically closest DXGW, with DXGW-to-CNE associations constrained to the same region for traffic isolation and latency optimization.

2. Scope

This technical analysis focuses on the Hybrid connectivity between Edge connects (San Jose, Ashburn, Dallas and Chicago)which are interconnected via full-mesh on-premise WAN backbone(Core) and AWS specifically the AWS. The testing of this design pattern is currently on-going in the Lab to further validate it feasibility in alignment to the design requirements of MCBB3.0

3.Architecture Overview

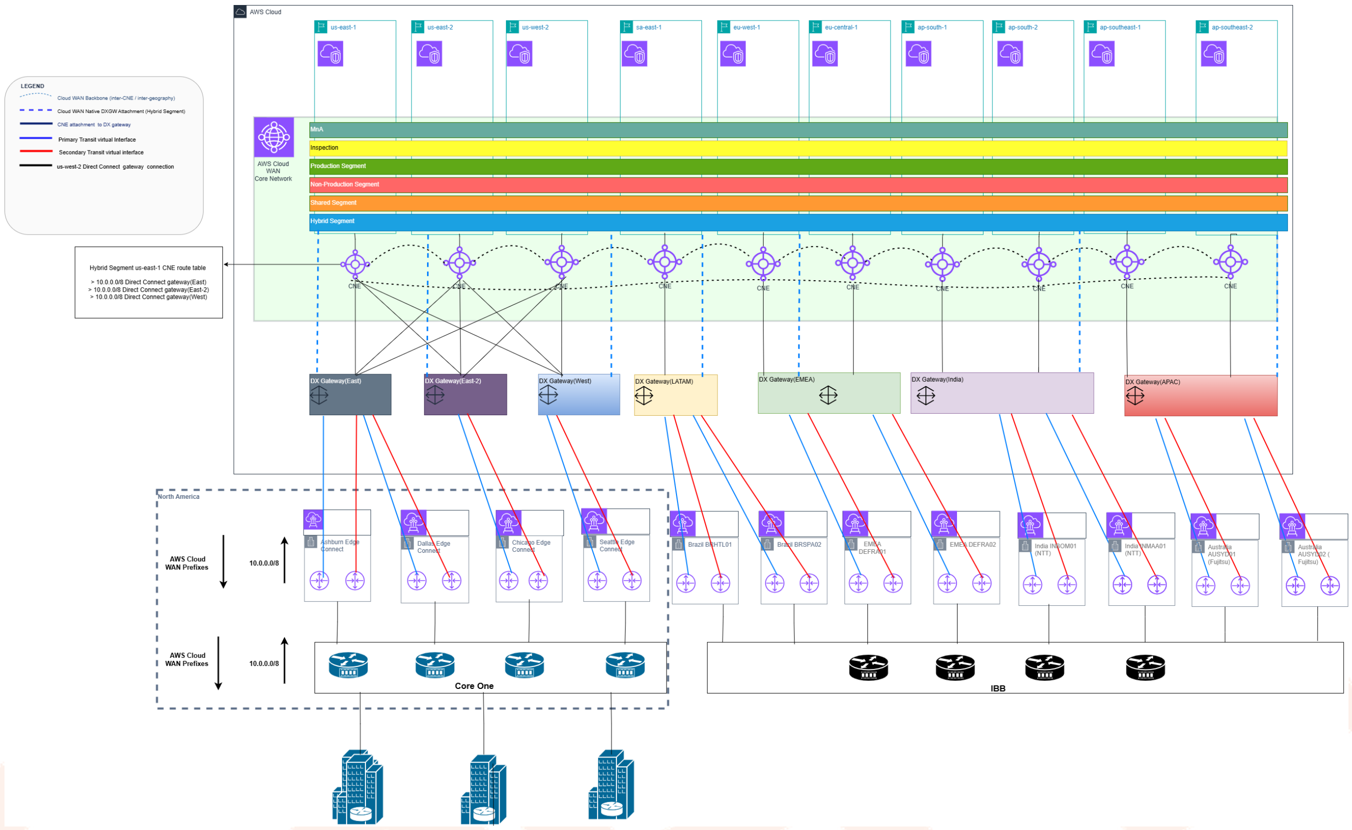
Seven DXGWs across North American and international regions to optimize hybrid connectivity and minimize latency.

North America (3 DXGWs):
One DXGW per AWS region, each supporting six VIFs from Edge Connects locations. All three North American DXGWs are associated with all three NA CNEs (us-east-1, us-east-2, us-west-2), creating a full-mesh topology. This architecture ensures every CNE maintains a direct local path to every DXGW, enabling any-to-any connectivity without traversing the Cloud WAN backbone. By eliminating cross-region backbone hops, traffic from Edge Connects can reach any North American region via the most optimal DXGW path.

International Regions (4 DXGWs):
One DXGW per geographical region with two VIFs per colocation/data center. Each regional DXGW is associated exclusively with CNEs in its respective region to maintain traffic locality and optimize latency:

- **LATAM DXGW** sa-east-1 (Sao Paulo) CNE
- **EMEA DXGW** eu-west-1 (Ireland) and eu-central-1 (Frankfurt) CNEs
- **India DXGW** ap-south-1 (Mumbai) and ap-south-2 (Hyderabad) CNEs
- **APAC DXGW** ap-southeast-1 (Singapore) and ap-southeast-2 (Sydney) CNEs

This regional isolation ensures international traffic remains within its geographic boundary for Direct Connect ingress, while inter-region communication traverses the Cloud WAN backbone with appropriate inspection and routing policies.



LEGEND
Cloud WAN Backbone (inter-CNE / inter-geography)
Cloud WAN Native DXGW Attachment (Hybrid Segment)
CNE attachment to DX gateway
Primary Transit virtual Interface
Secondary Transit virtual interface
us-west-2 Direct Connect gateway connection
Hybrid Segment us-east-1 CNE route table
> 10.0.0.0/8 Direct Connect gateway(East)
> 10.0.0.0/8 Direct Connect gateway(East-2)
> 10.0.0.0/8 Direct Connect gateway(West)
AWS Cloud
us-east-1
us-east-2
us-west-2
sa-east-1
eu-west-1
eu-central-1
ap-south-1
ap-south-2
ap-southeast-1
ap-southeast-2
AWS Cloud WAN Core Network
MnA
Inspection
Production Segment
Non-Production Segment
Shared Segment
Hybrid Segment
CNE
DX Gateway(East)
DX Gateway(East-2)
DX Gateway(West)
DX Gateway(LATAM)
DX Gateway(EMEA)
DX Gateway(India)
DX Gateway(APAC)
North America
AWS Cloud WAN Prefixes
10.0.0.0/8
Core One
IBB

3.2 Route Advertisement Behaviour

When a DXGW is attached to multiple CNE edge locations, AWS-documented behaviour defines how routes flow in both directions:

- Inbound (Edge Connects AWS): The DXGW receives 10.0.0.0/8 from all VIFs. It advertises these to all three associated CNEs. Each CNE installs the routes with the DXGW as next-hop and a short AS_PATH (local edge, direct attachment). and from the other CNEs in the Core Network with a longer AS_PATH.
- Outbound (AWS DC): Each CNE advertises ONLY its own local VPC/segment routes back to the DXGW — not routes learned from other CNEs. The DXGW aggregates local routes from all three CNEs and advertises the combined set to each Each Connects locations over their Transit VIF. The Edge connects re-advertise the aggregated route to CoreOne which route the destination packet to the On-premises DC.
- AS_PATH visibility: the full BGP AS_PATH is preserved end-to-end, so on-premises routers can see which CNE ASN originated each VPC prefix.

With all 3 DXGWs attached to all 3 CNEs, every CNE receives 10.0.0.0/8 from three separate DXGW attachments simultaneously, Longest Prefix Match cannot differentiate DXGWs , A Cloud WAN routing policy is REQUIRED to ensure each CNE prefers the geographically correct DXGW for it outbound traffic towards the Edge connect. For more advanced routing and traffic engineering, the recently released Advanced routing policy can be leveraged within the CNE .

3.3 Traffic Flow Deep-Dive: EC2 (us-east-1) Johns Creek DC

The target is a server at 10.10.44.10 located in the Johns Creek datacenter, reachable via Core One from the Ashburn or Dallas Edge colo.

AWS only knows 10.0.0.0/8. It cannot distinguish 10.10.44.10 (Johns Creek) from 10.20.55.20 (Omaha) at the AWS routing layer. Both destinations resolve to the same /8 prefix on all three DXGWs. The Cloud WAN routing policy must select the preferred DXGW for each CNE. Once traffic exits AWS at an Edge colo, Core One routes to the correct DC internally based on its own routing table.

3.3.1 Cloud WAN Routing Policy — DXGW Preference per CNE

The routing policy sets local preference on each CNE for each DXGW attachment. The design goal is to route each CNE's traffic to the geographically closest Edge connect:

- CNE us-east-1: prefers DXGW-EAST (Ashburn + Dallas Edge colos, lowest latency to East region)
- CNE us-east-2: prefers DXGW-EAST-2 (Chicago Edge colo, lowest latency to Central region)
- CNE us-west-2: prefers DXGW-WEST (San Jose Edge colo, lowest latency to West region)
- Fallback: lower LP to non-preferred DXGWs for resilience

3.3.2 Step-by-Step: EC2 (us-east-1) Johns Creek DC (10.10.44.10)

Egress path (AWS Johns Creek):

- Step 1: EC2 (10.1.5.20, us-east-1 VPC) sends packet to 10.10.44.10. VPC route table forwards to CNE-USE1 via the Cloud WAN attachment (NA-PROD segment).
- Step 2: CNE-USE1 Hybrid segment route table lookup. Three routes for 10.0.0.0/8: via DXGW-EAST (LP=300, policy-set), via DXGW-EAST-2 (LP=200), via DXGW-WEST (LP=100). Best path: DXGW-EAST (highest LP). All DXGWs are local to CNE-USE1 (all-CNE attachment). No backbone hop.
- Step 3: CNE-USE1 forwards to DXGW-EAST. DXGW-EAST BGP table has four VIFs: Ashburn VIF-1, Ashburn VIF-2, Dallas VIF-1, Dallas VIF-2 — all advertising 10.0.0.0/8. Equal attributes — DXGW-EAST ECMP load-balances across all four active VIFs per-flow.
- Step 4: Packet exits via one of the Ashburn or Dallas Edge colo DX connections. Assume Ashburn VIF-1 is selected.

- Step 5: Packet arrives at Ashburn Edge colo. Core One receives it and looks up 10.10.44.10 in its routing table. Core One forwards to Johns Creek DC via its internal network fabric.
- Step 6: Johns Creek DC server receives packet at 10.10.44.10.

Ingress path (Johns Creek EC2):

- Johns Creek DC sends return packet to 10.1.5.20. Core One routes to Ashburn Edge colo (or Dallas Edge colo — Core One selects based on its own routing policy). Ashburn Edge colo sends over DX DXGW-EAST.
- DXGW-EAST looks up 10.1.0.0/16 (VPC-A CIDR). DXGW-EAST is attached to all NA CNEs, so it has received 10.1.0.0/16 from CNE-USE1. Next-hop: CNE-USE1. DXGW-EAST forwards to CNE-USE1 VPC-A EC2.

Failover — Ashburn VIF-1 physical port failure:

- Ashburn VIF-1 drops. DXGW-EAST still has Ashburn VIF-2, Dallas VIF-1, Dallas VIF-2 active. DXGW-EAST immediately reroutes to remaining VIFs (BFD sub-second). No change at CNE level — DXGW-EAST still preferred by CNE-USE1 routing policy.

Failover — DXGW-EAST failure:

- DXGW-EAST Cloud WAN attachment goes down. CNE-USE1 loses LP=300 path. Falls back to DXGW-EAST-2 (LP=200). Traffic exits via Chicago Edge colo Core One Johns Creek DC. Latency increases slightly (Chicago Johns Creek via Core One) but connectivity maintained.